

Auteur: Michael Zielinski
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$$\int_0^1 x(1-x)^8 dx$$

$$u = 1 - x$$

$$du = -dx$$

$$x = 1 - u$$

$$x = 0 \Rightarrow u = 1$$

$$x = 1 \Rightarrow u = 0$$

$$\begin{aligned} \int_0^1 x(1-x)^8 dx &= \int_1^0 (1-u)u^8(-du) \\ &= \int_0^1 (1-u)u^8 du \\ &= \int_0^1 (u^8 - u^9) du \\ &= \left[\frac{u^9}{9} - \frac{u^{10}}{10} \right]_0^1 \\ &= \left(\frac{1}{9} - \frac{1}{10} \right) - 0 \\ &= \frac{10}{90} - \frac{9}{90} \\ &= \frac{1}{90} \end{aligned}$$

References